

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 4

Date Due: See canvas

This assignment has a total of **(145 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 4.1 (20 pts) Let X be binomial with $n = 20$ and $\theta = 0.3$. Use the normal approximation to approximate each of the following.

- (i) (5 pts) $\Pr[X \leq 3]$.
- (ii) (5 pts) $\Pr[3 \leq X \leq 6]$.
- (iii) (5 pts) $\Pr[X \geq 4]$.
- (iv) (5 pts) $\Pr[X = 4]$.

Exercise 4.2 (20 pts) Opponents of a nuclear power project claim that the majority of those living near a proposed site are opposed to the project. To justify this statement, a random sample of 75 residents is selected and their opinions are sought. Let X denote the number opposed to the project.

- (i) (5 pts) If the probability that an individual is opposed to the project is 0.5, is the normal approximation appropriate?
- (ii) (5 pts) If $\theta = 0.5$, what is $\mathbb{E}[X]$?
- (iii) (5 pts) If $\theta > 0.5$ as claimed, then, on the average, more than 37.5 of every 75 individuals are opposed to the project. Let us agree to accept the claim if X is at least 46. What is the probability that we shall accept the claim if θ is really only 0.5?
- (iv) (5 pts) What is the probability that we shall not accept the claim ($X < 45$) even though it is true and θ is really 0.7?

Exercise 4.3 (35 pts) Computer chips do not “wear out” in the ordinary sense. Assuming that defective chips have been removed from the market by factory inspection, it is reasonable to assume that these chips exhibit a constant hazard rate. Let the hazard rate be given by $h(t) = 0.02$. (Time is in years.)

- (i) (5 pts) In a practical sense, what are the main causes of failure of these chips?
- (ii) (5 pts) What is the reliability function for chips of this type?
- (iii) (5 pts) What is the reliability of a chip 20 years after it has been put into use?
- (iv) (5 pts) What is the failure density for these chips?
- (v) (5 pts) What type of random variable is X , the time to failure of a chip?
- (vi) (5 pts) What is the mean and variance for X ?
- (vii) (5 pts) What is the probability that a chip will be operable for at least 30 years?

Exercise 4.4 (10 pts) Let X denote the velocity of a random gas molecule. According to the Maxwell-Boltzmann law, the density for X is given by

$$f_X(x) = cx^2 e^{-\beta x^2}, \quad x > 0$$

Here c is a constant that depends on the gas involved, and β is a constant whose value depends on the mass of the molecule and its absolute temperature. The kinetic energy of the molecule, Y , is given by $Y = \frac{1}{2}mX^2$ where $m > 0$. Find the density for Y .

Exercise 4.5 (50 pts) An engineer is studying early morning traffic patterns at a particular intersection. The observation period begins at 5:30 a.m. Let X denote the time of arrival of the first vehicle from the north-south direction; let Y denote the first arrival time from the east-west direction. Time is measured in fractions of an hour after 5:30 a.m. Assume that the density for (X, Y) is given by

$$f_{XY}(x, y) = 1/x, \quad 0 < y < x < 1$$

- (i) (5 pts) Verify that this is a joint density for a two-dimensional random variable.

- (ii) (5 pts) Find $\Pr[X \leq 0.5, Y \leq 0.25]$.
- (iii) (5 pts) Find $\Pr[X > 0.5 \text{ or } Y > 0.25]$.
- (iv) (5 pts) Find $\Pr[X \geq 0.5, Y \geq 0.5]$
- (v) (5 pts) Find the marginal densities for X and Y .
- (vi) (5 pts) Find $\Pr[X \leq 0.5]$.
- (vii) (5 pts) Find $\Pr[Y \leq 0.25]$.
- (viii) (5 pts) Are X and Y independent? Explain.
- (ix) (10 pts) Find $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, and $\text{Cov}(X, Y)$.

Exercise 4.6 (10 pts) Given rv's N and X . Suppose the rv $N \sim \text{Poisson}(\lambda)$, and $X \mid (N = n) \sim \text{Binomial}(n, p)$. Show that $X \sim \text{Poisson}(p\lambda)$.